

Material volume vs. Control volume

Reynolds Transport Theorem

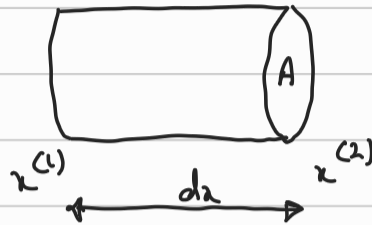
Material volume vs. Control volume:

$$x = \hat{x}(X, t)$$

$$y = Y$$

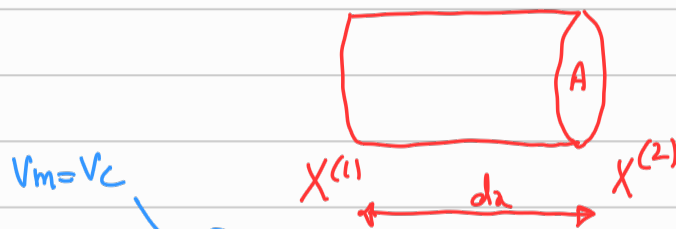
$$z = Z$$

$$\rho = \rho(x, t)$$



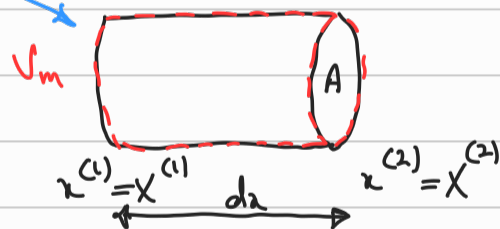
V_C : Control volume

$$m(x^{(1)}, x^{(2)}, t) = \int_{x^{(1)}}^{x^{(2)}} \rho(x, t) A dx \quad \text{integral over control volume}$$



V_m : material volume

At an instance when $V_m = V_C$, we would like to calculate mass (M)



$$M(X^{(1)}, X^{(2)}, t) = \int_{x^{(1)}(X^{(1)}, t)}^{x^{(2)}(X^{(2)}, t)} \rho(x, t) A dx$$

$$\frac{\partial m}{\partial t} = \frac{\partial}{\partial t} \left[\int_{x^{(1)}}^{x^{(2)}} \rho(x, t) A dx \right]_{x^{(1)}, x^{(2)} \text{ fixed}}$$

the rate of
at which mass
is increasing
inside a fixed
control volume

$$\frac{\partial M}{\partial t} = \frac{\partial}{\partial t} \left[\int_{x(X^{(1)}, t)}^{x(X^{(2)}, t)} \rho(x, t) A dx \right]_{X_1, X_2 \text{-fixed}}$$

$$= \frac{D}{Dt} \int_{x(X^{(1)}, t)}^{x(X^{(2)}, t)} \rho(x, t) A dx$$

the rate of increase
of mass of that
specific part of material
that coincides the fixed
control volume at time t

$$V_C: \text{control volume} \quad \int_{V_C} T(x, t) dV \xrightarrow{\partial/\partial t} \frac{\partial}{\partial t} \int_{V_C} T(x, t) dV$$

$$V_m: \text{material volume} \quad \int_{V_m} T(x, t) dV \xrightarrow{\partial/\partial t} \frac{D}{Dt} \int_{V_m} T(x, t) dV$$

Raynolds Transport Theorem:

$$\begin{aligned} \frac{D}{Dt} \int_{V_m} T(x, t) dV &= \int_{V_C} \frac{\partial T(x, t)}{\partial t} dV + \int_{S_C} T(r \cdot n) dS \\ &= \int_{V_C} \frac{DT(x, t)}{Dt} dV + \int_{V_C} T(\text{div } v) dV \end{aligned}$$

Proof (1):

$$\frac{D}{Dt} \int_{V_m(t)} T(x, t) dV = \int_{V_m=V_C} \frac{D}{Dt} (T dV) = \int_{V_C} \frac{DT}{Dt} dV + T \frac{D(dV)}{Dt}$$

$$\frac{D(dV)}{Dt} = (\text{div } v) dV \rightarrow \int_{V_C} \frac{DT}{Dt} dV + T(\text{div } v) dV$$

$$= \int_{V_c} \left(\frac{DT}{Dt} + T(\operatorname{div} v) \right) dV$$

Cartesian

Coordinate : $\frac{D}{Dt} \int_{V_m(t)} T_{ij} dV = \int_{V_c} \left(\frac{DT_{ij}}{Dt} + T_{ij} \frac{\partial v_k}{\partial x_k} \right) dV$

System

$$\begin{aligned} \frac{D(\cdot)}{Dt} &= \frac{\partial(\cdot)}{\partial t} + (\nabla(\cdot)) \cdot v \\ \hline &\rightarrow \int_{V_c} \left(\frac{\partial T_{ij}}{\partial t} + \frac{\partial T_{ij}}{\partial x_k} v_k + T_{ij} \frac{\partial v_k}{\partial x_k} \right) dV \\ &= \int_{V_c} \left\{ \frac{\partial T_{ij}}{\partial t} + \frac{\partial (T_{ij} v_k)}{\partial x_k} \right\} dV \\ &= \int_{V_c} \frac{\partial T_{ij}}{\partial t} dV + \int_{S_c} T_{ij} v_k n_k dS \end{aligned}$$

$$\frac{D}{Dt} \int_{V_m} T dV = \int_{V_c} \frac{\partial T}{\partial t} dV + \int_{S_c} T(v \cdot n) dS$$

Proof (2):

$$dV = J dV_0 \rightarrow dV = (\det F) dV_0$$

$$\int_{V_m} T dV = \int_{V_0} T (\det F) dV_0 \xrightarrow{\partial/\partial t}$$

$$\frac{D}{Dt} \int_{V_m} T dV = \frac{D}{Dt} \int_{V_0} T (\det F) dV_0$$

$$= \int_{V_0} \frac{D}{Dt} (T (\det F)) dV_0$$

$$= \int_{V_0} \left\{ \frac{DT}{Dt} (\det F) + T \frac{D(\det F)}{Dt} \right\} dV_0$$

$$\begin{cases} \frac{D(dv)}{Dt} = (\operatorname{div} v) dv \\ dv = (\det F) dV_0 \end{cases} \Rightarrow \frac{D}{Dt} ((\det F) dV_0) = \frac{D(\det F)}{Dt} dV_0 + \cancel{\det F \frac{D(dv)}{Dt}}$$

$$\Rightarrow \frac{D(\det F)}{Dt} = \frac{1}{dV_0} \frac{D}{Dt} ((\det F) dV_0) \quad \leftarrow \frac{D(dv)}{Dt}$$

$$= \frac{1}{dV_0} \frac{D(dv)}{Dt}$$

$$= \frac{1}{dV_0} (\operatorname{div} v) (dv)$$

$$= \frac{1}{\cancel{dV_0}} (\operatorname{div} v) (\det F) \cancel{dV_0}$$

$$= (\operatorname{div} v) (\det F)$$

$$= \int_{V_0} \left\{ \frac{DT}{Dt} (\det F) + T \frac{D(\det F)}{Dt} \right\} dV_0$$

$$= \int_{V_0} \left\{ \frac{DT}{Dt} (\det F) + T (\operatorname{div} v) (\det F) \right\} dV_0$$

$$= \int_{V_0} \left\{ \frac{DT}{Dt} + T (\operatorname{div} v) \right\} \cancel{\det F} \overset{\swarrow}{dV_0} \quad \leftarrow \frac{D(dv)}{Dt}$$

$$= \int_{V_0} \left\{ \frac{DT}{Dt} + T (\operatorname{div} v) \right\} dv$$